

8.6 Calling GitHub Copilot and ChatGPT in the Terminal

If you work a lot in the terminal, it makes sense to call ChatGPT and others directly from there. There are various ways of doing this, two of which are described here: `gh` and `sgpt`.

8.6.1 GitHub Command-Line Interface (“gh”)

The `gh` command enables you to carry out various GitHub actions directly in the terminal. `gh` is therefore a *command-line interface* (CLI) to the GitHub web interface. Installation instructions for Windows, Linux, and macOS can be found at <https://github.com/cli/cli>.

You must carry out authentication with `gh auth` before using it for the first time. Finally, you must install the `gh` extension, `gh-copilot`:

```
gh extension install github/gh-copilot
```

Assuming that you have a GitHub Copilot subscription, you can then call two Copilot commands:

- `gh copilot explain "command"` provides explanations of the specified command.
- `gh copilot suggest "question"` suggests a solution for the specified question. Using `-t shell|gh|git` you can explicitly refer to terminal commands (Bash, Z shell [Zsh], PowerShell), to the `gh` command or to Git problems. In the course of further steps, `gh` can explain the suggested command, copy it to the clipboard, or execute it.

The following examples show how you can use the commands:

```
gh copilot explain "sed 5,7d < file"
```

Explanation:

- * sed is a command-line tool for text manipulation.
- * 5,7d specifies that we want to delete lines 5 to 7 from the input.
- * < file redirects the contents of the file to the input of the sed command.

```
gh copilot explain "Get-Process | Select-Object -First 5"
```

- * Get-Process retrieves a list of running processes.
- * Select-Object -First 5 selects the first 5 processes from the list.

```
gh copilot suggest -t shell "count IP addresses in a log file"
```

Suggestion:

```
grep -oE "\b([0-9]{1,3}\.){3}[0-9]{1,3}\b" logfile.txt | \
  sort | uniq -c | sort -nr
```

? Select an option

> Explain command

- * grep -oE "\b([0-9]{1,3}\.){3}[0-9]{1,3}\b" searches for IP addresses in the logfile.txt file ...

8.6.2 Shell-GPT (“sgpt”)

ChatGPT is only intended for use in web browsers. In the past, there have been tools that have tried to circumvent this. However, these no longer work today. The use of ChatGPT in the terminal therefore requires that you have paid API access to OpenAI (see <https://openai.com/api>). A ChatGPT subscription isn’t enough!

Based on this condition, you have the choice between various commands that can be run in the terminal. These commands send API requests to OpenAI and then display the response. Each call costs a little less than a cent, depending on the length of the answer. In mid-2024, OpenAI charged \$10 for one million output tokens generated with GPT-4o. This corresponds to a text volume of approximately 800 book pages.

We've tried sgpt. Installation instructions for Linux, macOS and Windows can be found at <https://github.com/tbckr/sgpt>.

The command requires the OPENAI_API_KEY environment variable to be defined. On Linux or macOS, you simply need to include the following statement in .bashrc or .zshrc. You must log out and log in again for the variable declaration to take effect.

```
# in the .bashrc (Linux) or .zshrc (macOS) file
...
export OPENAI_API_KEY='sk-Ke...'
```

The following listing shows three application examples. The last example is particularly interesting, where we asked sgpt to process the very confusing result of the whois command from the standard input in such a way that it's easier to understand (from a human perspective).

```
sgpt "is 172.16.5.240 in 172.16.5.160/28"
```

```
No, 172.16.5.240 is not in the range 172.16.5.160/28.
```

```
The network address 172.16.5.160 with a subnet mask of /28
means that the range of IP addresses includes 172.16.5.161 to
172.16.5.174.
```

```
sgpt "bash: how do I replace ; by : in a text file"
```

```
You can use the `sed` command in the terminal to replace `;`
by `:` in a text file. Here's an example command:
```

```
sed -i 's/;/:/g' file.txt
```

```
whois kofler.info | sgpt stdin "Summarize human readable"
```

```
The domain kofler.info is registered to Michael Kofler. The
registrar is Key-Systems GmbH and the domain was created on
September 26, 2003 and expires ...
```

8.6.3 Using Language Models in the Terminal (“llm”)

The `sgpt` command just described works well, but is specifically designed for the OpenAI API. The Python program `llm` takes a much more general approach. It supports various APIs and can use both network and local language models. Using `llm`, you can change the language model at any time.

Note

For more information on `llm`, take a look at these websites:

- <https://github.com/simonw/llm>
- <https://simonwillison.net/2024/Jun/17/cli-language-models>

It's best to use `pip` or `brew` (macOS) to install the Python script. Various libraries and modules are also downloaded in this process. If you use `pip`, you should first set up a directory with a Python environment and activate it. In our tests, the `pip` installation on Linux only worked on x86 hardware, but not if using an ARM CPU. (The errors don't occur during installation, but only later when the `llm` command gets used.)

```
pip[3] install llm
brew install llm
```

The next step is to install the `gpt4all` plugin, which allows `llm` to communicate with various language models (including the OpenAI API):

```
llm install llm-gpt4all
llm models list
```

```
OpenAI Chat: gpt-4o (aliases: 4o)
OpenAI Chat: gpt-4o-mini (aliases: 4o-mini)
...
gpt4all: Meta-Llama-3 - Llama 3.1 8B Instruct 128k,
        4.34GB download, needs 8GB RAM
gpt4all: orca-mini-3b-gguf2-q4_0 - Mini Orca (Small),
        1.84GB download, needs 4GB RAM
gpt4all: Phi-3-mini-4k-instruct - Phi-3 Mini Instruct,
        2.03GB download, needs 4GB RAM
```

```
gpt4all: wizardlm-13b-v1 - Wizard v1.2,  
6.86GB download, needs 16GB RAM
```

If `llm` is to use the OpenAI API, you should install the corresponding plugin in the next step and enter your API key:

```
llm keys set openai
```

```
Enter key: *****
```

You can then use `llm` to send prompts to the OpenAI API. The answer gets displayed in the terminal:

```
llm 'explain recursion'
```

```
Recursion is a programming and mathematical concept where a  
function calls itself directly or indirectly to solve a  
problem. It's a powerful technique that can simplify code and  
make certain problems easier to solve ...
```

Provided you have powerful hardware, you can also process prompts using a local language model. Prior to that, the model should be downloaded and saved in the `.cache/gpt4all` directory.

```
llm -m orca-mini-3b-gguf2-q4_0 'explain recursion'
```

```
Recursion is a programming technique ...
```

We'll leave it at that for now. You can find lots of tips on using `llm` on the GitHub page we gave the URL for earlier. In addition, we deal with the local execution of language models in detail in this book from [Chapter 9](#) onward.